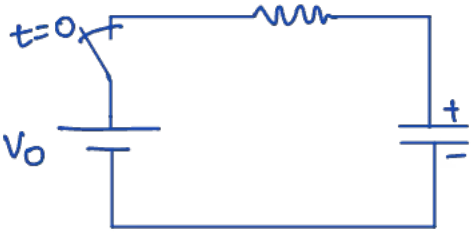
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Q1 → 20% Q2 → 20% EndSem → 50% Tutorials → 10%

① Sheegaokar ② Sadiku

Electric Field ($\vec{E}$): N/A.s or V/m



$$V_0 = i(t) \cdot R + V_c(t)$$

$$V_c(t) = \frac{C\, dV_c(t)}{dt} \cdot R + V_c(t)$$

$$\Rightarrow \frac{V_0 e^{t/RC}}{RC} = \frac{dV_c}{dt} \cdot e^{t/RC} + \frac{V_c \cdot e^{t/RC}}{RC}$$

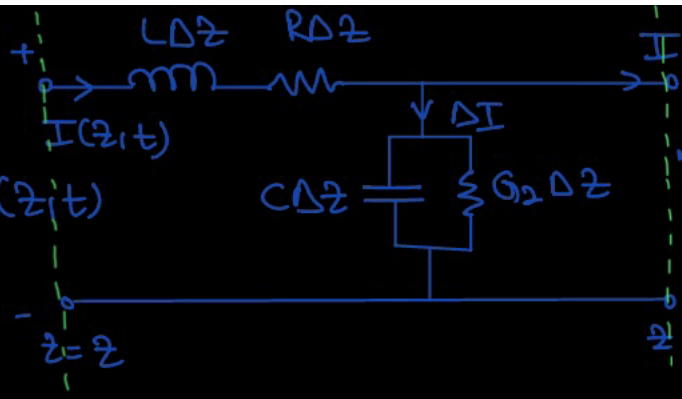
$$= \frac{d}{dt}\left(V_c \cdot e^{t/RC}\right)$$

$$\Rightarrow V_0 e^{t/RC} - V_0 = V_c e^{t/RC}$$

$$\Rightarrow V_c = V_0\left(1 - e^{-t/RC}\right)$$



After connecting battery, bulb doesn't glow immediately due to resistance offered by wires. We can divide them into smaller segments



R, L, C, G
↓
per unit length

$$V(z,t) = L\Delta z\frac{dI(z,t)}{dt} + I(z,t)\cdot R\cdot\Delta z + V(z+\Delta z,t)$$

$$\Rightarrow -\frac{V(z+\Delta z,t)-V(z,t)}{\Delta z} = RI(z,t) + L\frac{dI(z,t)}{dt}$$

$$\Rightarrow \boxed{-\frac{dV(z,t)}{dz} = R\cdot I(z,t) + L\cdot\frac{dI(z,t)}{dt}}$$

$$I(z,t) = I(z+\Delta z,t) + \Delta I$$

$$= I(z+\Delta z,t) + C\Delta z\frac{dV(z+\Delta z,t)}{dt} + G\Delta z\,V(z+\Delta z,t)$$

displacement current — conduction current

$$\Rightarrow \boxed{-\frac{dI(z,t)}{dz} = C\frac{dV(z,t)}{dt} + GV(z,t)}$$

we generally analyze transmission lines with voltage and current rather than electric/magnetic fields.

Assume $v(z,t) = Re[V_s(z) e^{j\omega t}]$

$I(z,t) = Re[I_s(z) e^{j\omega t}]$

We need to find out $V_s(z)$ and $I_s(z)$:

$$\Rightarrow -\frac{dV_s(z)}{dz} = (R + j\omega L) I_s(z)$$

$$\Rightarrow -\frac{dI_s(z)}{dz} = (G + j\omega C) V_s(z)$$

$$\Rightarrow \frac{d^2V_s(z)}{dz^2} = \underbrace{(G+j\omega C)(R+j\omega L)}_{\gamma^2} V_s(z) \quad ; \gamma = \alpha + i\beta$$

$$\Rightarrow \frac{d^2I_s(z)}{dz^2} = (G+j\omega C)(R+j\omega L) I_s(z)$$

$$\frac{d^2V_s(z)}{dz^2} = \gamma^2 V_s(z) \quad ; \quad \frac{d^2I_s(z)}{dz^2} = \gamma^2 I_s(z)$$

$$\Rightarrow V_s(z) = Ae^{\gamma z} + Be^{-\gamma z} \qquad A = V_0^- \; ; \; B = V_0^+$$

$$I_s(z) = Ce^{\gamma z} + De^{-\gamma z} \qquad C = I_0^- \; ; \; D = I_0^+$$

$$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

V_0^+ → amplitude of forward propogating voltage signal

I_0^+ → amplitude of forward propogating current signal

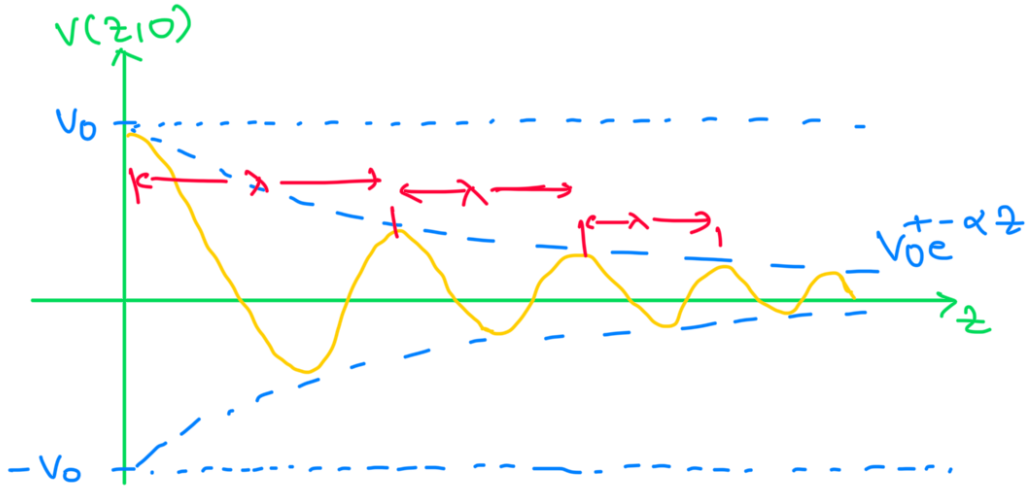
V_0^+ → amplitude of backward propogating voltage signal

I_0^+ → amplitude of backward propogating current signal

Substitute in $-\frac{dV_S}{dz} = (R+j\omega L) I_S$

we get $\frac{V_0^+}{I_0^+} = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$ $\frac{V_0^-}{I_0^-} = -\sqrt{\frac{R+j\omega L}{G+j\omega C}}$

$$V(z,t) = e^{-\alpha t} V_0^+ \cos(\omega t - \beta z) + V_0 e^{+\alpha t} \cos(\omega t + \beta z)$$



$\beta . \lambda = 2\pi \Rightarrow \beta = \frac{2\pi}{\lambda} \Rightarrow v_p = \omega/\beta$

$\omega = \frac{2\pi v_p}{\lambda} = \beta v_p$ v_p → phase velocity

$\omega t_1 - \beta z_1 = \omega t_2 - \beta z_2 \Rightarrow \frac{\omega}{\beta} = \frac{z_2 - z_1}{t_2 - t_1}$

z_0 (impedance) $= \frac{V_0^+}{} = \sqrt{\frac{R+j\omega L}{}}$ ohm

I_0^+ $\int$ $G + j\omega C$

$Z_0 = R_0 + jX_0$ $X_0 \rightarrow$ characteristic

impedance

Ideal Transmission Line:

$R = 0$ $\Rightarrow \gamma = j\omega\sqrt{LC} = \alpha + i\beta$

$G = 0$

$\alpha = 0$, $\beta = \omega\sqrt{LC}$ $\Rightarrow \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$ (independent of frequency)

$L > \mu_0$
$C > \varepsilon_0$

$Z_0 = R_0 + jX_0 \Rightarrow \sqrt{\frac{L}{C}} = R_0 + jX_0 \Rightarrow R_0 = \sqrt{\frac{L}{C}}$; $X_0 = 0$

Transmission lines recap:-

$$V(z,t) = Re\left[V_s(z)\, e^{j\omega t}\right]$$

$$I(z,t) = Re\left[I_s(z)\, e^{j\omega t}\right]$$

$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$ $I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$

$\gamma = \alpha + i\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$ = propagation constant

$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$

$u_p = \frac{\omega}{\beta}$ ($u_p \rightarrow$ phase velocity)

In Reality:- $R \neq 0$; $G \neq 0$ but we don't want distortion

Considering all the external factors, design

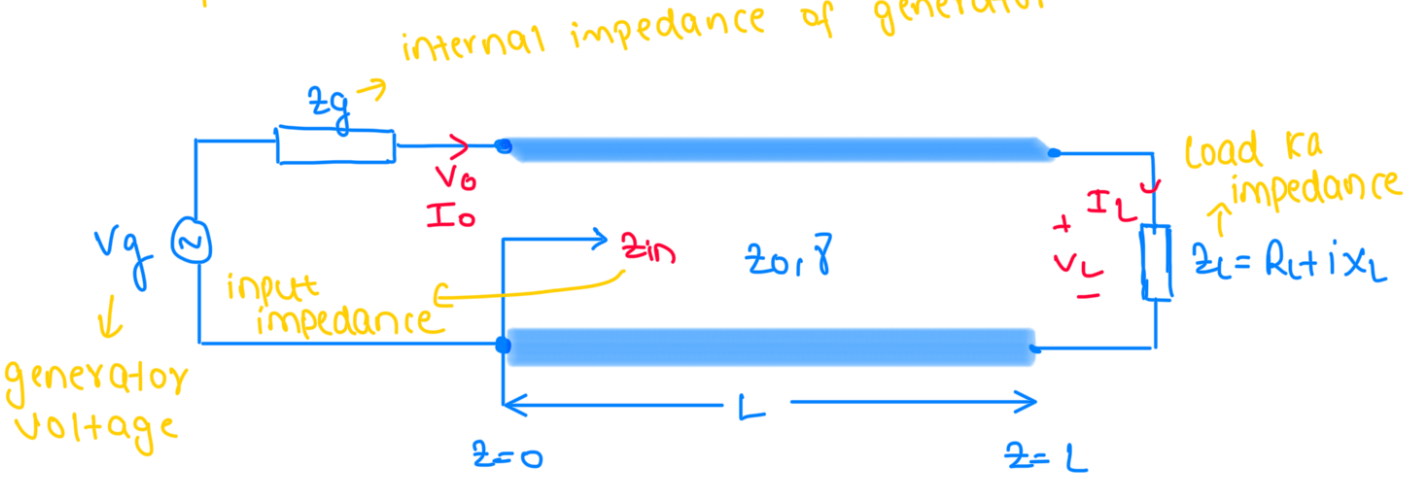
a transmission line with $R/L = G/C$

$\Rightarrow z_0 = \sqrt{L/C}$; $R_0 = \sqrt{L/C}$ $x_0 = 0$ (similar to ideal cases)

$$\gamma = \sqrt{RG}\left(1 + \frac{j\omega L}{R}\right) = \sqrt{RG}\left(1 + \frac{j\omega C}{G}\right)$$

$$\Rightarrow \alpha = \sqrt{RG} \quad \beta = \omega\sqrt{LC} \rightarrow \omega/\beta = 1/\sqrt{LC}$$

There will be losses ($\because R, G \neq 0$), but we can design a transmission line which is phase independent (Distortionless transmission line)



$$V_0 = V_0^+ + V_0^- \quad \Rightarrow V_0^+ = \frac{V_0 + I_0 z_0}{2}$$

$$I_0 z = V_0^+ - V_0^- \qquad V_0^- = \frac{V_0 - I_0 z_0}{2}$$

$V_L = V_s(z=L)$; $I_L = I_s(z=L)$

$$V_L = V_0^+ e^{-\gamma L} + V_0^- e^{\gamma L} \quad ; \quad I_L z = V_0^+ e^{-\gamma L} - V_0^- e^{-\gamma L}$$

$$V_0^+ = (V_L + I_L z_0) e^{\gamma L}/2$$

$$V_0^- = (V_L - I_L z_0) e^{-\gamma L}/2$$

$$z_{in} = \frac{V_s(z=0)}{I_s(z=0)} = z_0 \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} = z_0 \left[\frac{z_L + z_0 \tanh(\gamma L)}{z_0 + z_L \tanh(\gamma L)}\right]$$

Lossless TL: $\gamma = \alpha + j\beta = j\omega\sqrt{LC}$

$$Z_{in} = Z_0 \left\{ \frac{Z_L + Z_0 \cdot j \cdot \tan\beta l}{Z_0 + Z_L \cdot j \cdot \tan\beta l} \right\} = \text{complex always}$$

Γ_L^v (Voltage reflection coefficient at load)

$$= \frac{V_0^- e^{\gamma l}}{V_0^+ e^{-\gamma l}} = \frac{V_L - I_L Z_0}{V_L + I_L Z_0} = \frac{\text{Backward amp @ ld}}{\text{Forward amp @ ld}}$$

$$Z_L = V_L / I_L \Rightarrow \Gamma_L^v = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$\Gamma_L^v = 0 \Rightarrow Z_L = Z_0 \Rightarrow$ all power is delivered

Γ_L^e (current reflection coefficient at load)

$$= \frac{I_0^- e^{\gamma l}}{I_0^+ e^{-\gamma l}} = \frac{Z_0 - Z_L}{Z_0 + Z_L} = -\Gamma_L^v$$

When current reflects; we get additional π phase difference for current

VSWR → Voltage Standing wave Ratio

$$S = \frac{V_{max}}{V_{min}} = \frac{I_{max}}{I_{min}} = \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} = \frac{1 + \frac{V_0^-}{V_0^+}}{1 - \frac{V_0^-}{V_0^+}} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

Average power

$P_{avg} = \frac{1}{2} \text{Re}\{ V_s(l) I_s^*(l) \}$ → complex conductio

$$= \frac{1}{2} \text{Re}\left\{ V_0^+ \left[e^{-j\beta l} + \Gamma_L^0 e^{j\beta L} \right] \times \frac{V_0^{+*}}{z_0} \left[e^{j\beta L} - \Gamma_L^{0*} e^{-j\beta L} \right] \right\}$$

$$= \frac{1}{2} \text{Re}\left\{ \frac{|V_0^+|^2}{z_0} \left[1 - |\Gamma_L|^2 + \Gamma e^{-j\cdot 2\beta l} - \Gamma^* e^{j\cdot 2\beta l} \right] \right\}$$

(complex)

$$= \frac{1}{2} \left\{ \frac{|V_0^+|^2}{z_0} \left[1 - |\Gamma_L|^2 \right] \right\}$$

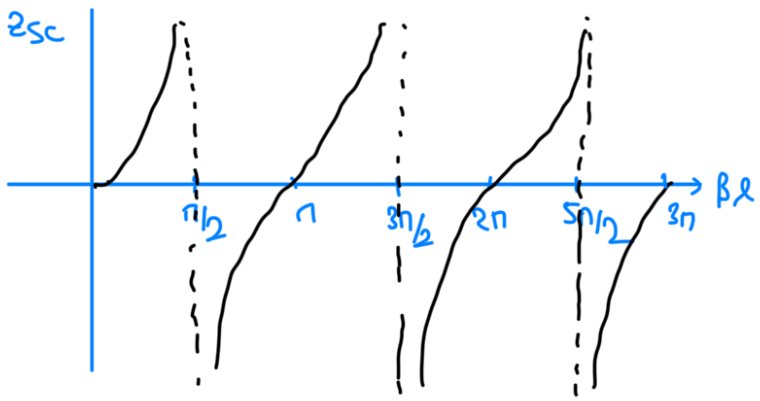
↓ power delivered

↓ power going fwd

→ power reflected

Shorted Transmission line: $z_L = 0$

$$z_{in} = z_{sc} = j z_0 \tan(\beta l)$$



$z_L = j\omega L$

$z_c = \frac{1}{j\omega C}$

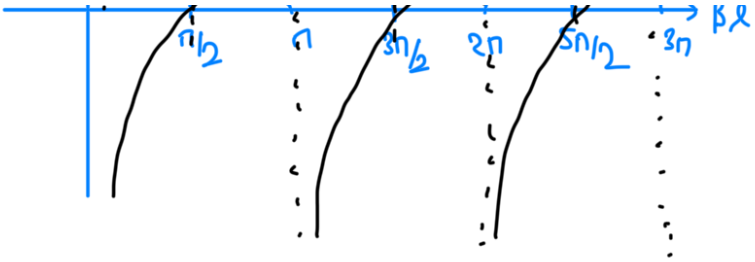
$\Gamma_L = -1$; $S = \infty$

Open transmission line $z_L \to \infty$

$$z_{in} = z_{oc} = \frac{z_0}{j\tan(\beta l)} = -j z_0 \cot(\beta l)$$



$\Gamma_L = 1$; $S = \infty$



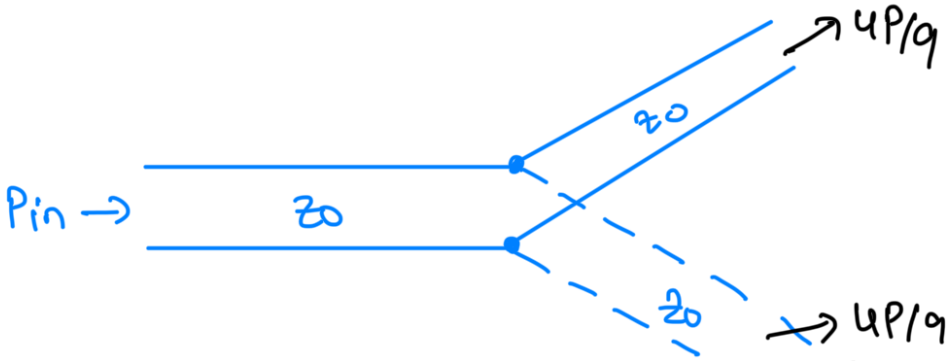
$$Z_{oc}\, Z_{sc} = Z_0^2$$

Matched Transmission line: $Z_L = Z_0$

$Z_m = Z_0$; $\Gamma_L = 0 \Rightarrow S = 1$

$\beta\ell$: $\beta\ell = \pi/2 \Rightarrow Z_{in} = Z_0'^2/Z_L$ (designed by)

$\ell = \lambda/4$ $\quad Z_{in} = Z_0 \Rightarrow Z_0' = \sqrt{Z_0 Z_L}$



$Z_L = Z_0/2 \Rightarrow \Gamma_L = -\frac{1}{3} \Rightarrow P_{delivered} = P_{in}(1 - |\Gamma_L|^2) = \frac{8P_{in}}{9}$

$Z_0' = \sqrt{Z_0 \times Z_0/2} = Z_0/\sqrt{2}$ for all power to be delivered

$$\text{loss (dB)} = 10 \log\left(\frac{P_{out}(\ell)}{P_{in}(0)}\right) = 20 \log\left(\frac{V_{out}(\ell)}{V_{in}(0)}\right)$$

$P_{out} = P_{in} e^{-2\alpha\ell} \Rightarrow \text{loss (dB)} = 8.68\, \alpha\, \ell \rightarrow m$ (α ↓ Np/m)

Return loss: $20 \log\left\{\frac{V_0^{\ominus}(z=0)}{V_0^{\oplus}(z=0)}\right\}$

Near end Cross Talk:

Far end Cross Talk..